IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-511

1. Topic (12 words max)

Smart Safety Monitoring System For Long-Distance Buses In Sri Lanka

2. Research group the project belongs to

CEAI – Centre of Excellence in AI

3. Specialization of the project belongs to

Software Engineering

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In Sri Lanka, many serious bus accidents happen every year, mostly on long-distance and night routes. These accidents are caused by things like driver tiredness, speeding, carrying too many passengers, and not having good emergency systems. Our safety system is important because it tries to fix these problems using easy and cheap technology.

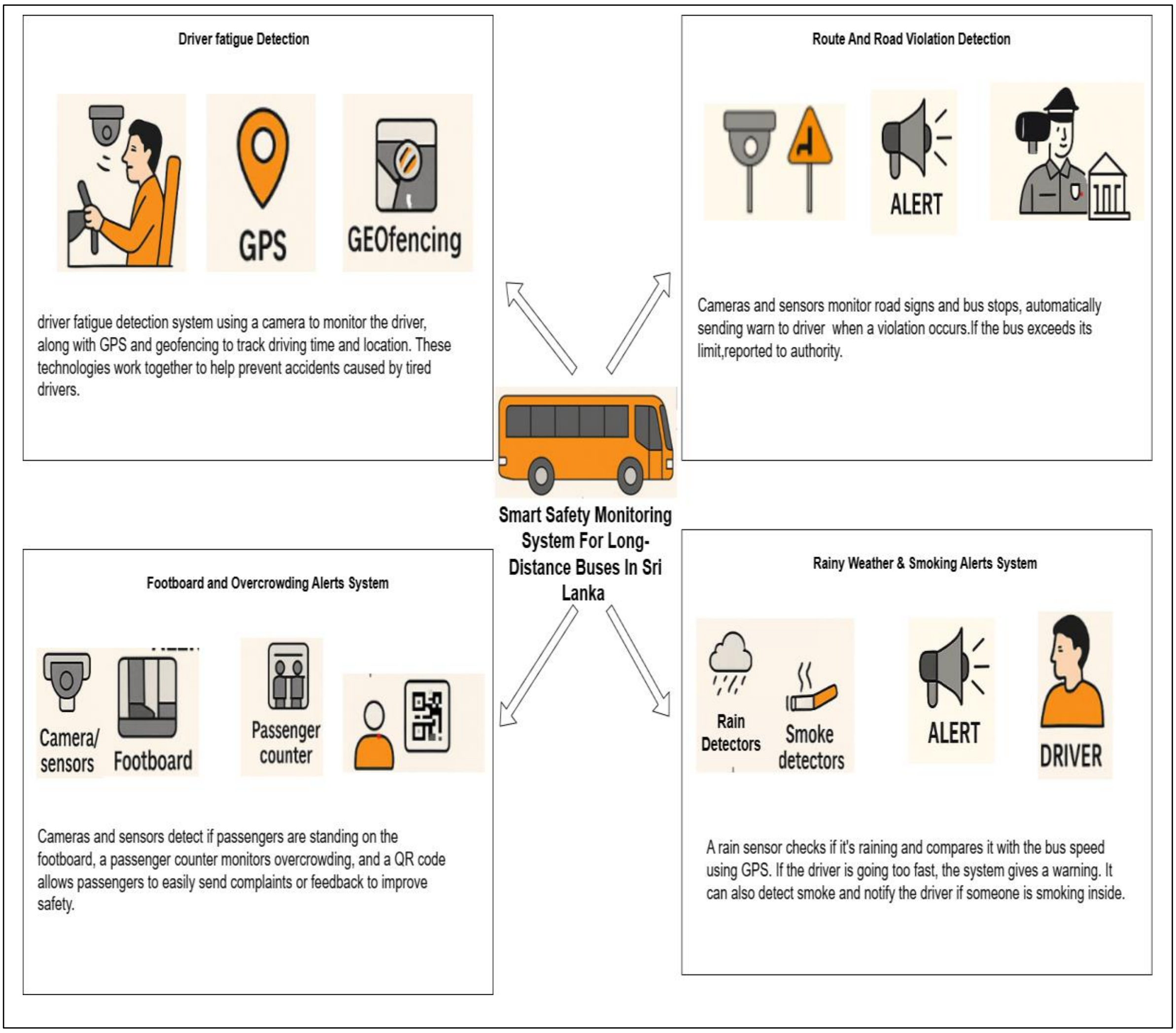
One big problem is that bus drivers drive for too long without taking a rest. According to SLTB rules, a driver should not drive more than six hours in a row. If the trip is longer, there should be a new driver or a different bus. But in real life, this rule is not always followed because there is no good way to check how long a driver has been driving. Also, drivers get sleepy during long trips. When they are tired, their eyes may close for some seconds, they may yawn a lot, or their heads may nod. These are clear signs the driver is too tired to drive safely. But right now, there is no system that can detect these signs and warn before something bad happens.

Also, in Sri Lanka, there are only a few places where authorities check if buses follow traffic rules. These checkpoints only check some vehicles at some points, not all the time or during the whole trip. So, many buses break rules without getting caught. For example, some buses skip their scheduled stops, so passengers miss their chance to get on or off. Some buses stay too long at stops just to waste time, which causes traffic jams and delays other buses. Sometimes, buses drop passengers in unsafe or wrong places, which is dangerous. But there is no system to watch the bus route and stops for the whole trip. This makes managing buses hard and causes problems for passengers and other road users.

Another problem is that many passengers, especially school kids, ride on the bus footboard even when there are empty seats. This is very unsafe because standing on the footboard can cause falls or injuries, especially when the bus moves suddenly. Also, some drivers drive too fast or let too many passengers ride. But passengers don't have a way to complain or warn drivers about these dangers. There is no system on buses to detect if someone stays on the footboard too long or to alert the driver to slow down or control passengers. Without this, passengers stay at risk.

Finally, many bus drivers continue to drive fast during rain, even though wet roads and low visibility increase the risk of accidents. Currently, no system advises them to slow down in such conditions. Additionally, some passengers smoke inside the bus, creating health risks and discomfort, but there's no way to detect or report it. To address these safety concerns, a system that can detect rain, predict bad weather, sense smoking, and alert the driver is essential for protecting everyone on board.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



Our Bus Passenger Safety System aims to reduce accidents and improve safety in Sri Lankan buses using simple, low-cost technology. It has four main features.

1. **Driver Fatigue Detection:** A camera checks the driver's eyes, mouth, and posture for signs of sleepiness like yawning, nodding, or closed eyes. It also tracks how long the driver has been on duty using GPS and time data. If the driver looks tired or drives over 6 hours without rest, the system warns the driver and sends alerts to the bus company or SLTB.

2. **Route and Road Violation Detection:** A GPS tracker and geofencing ensure buses stop only at approved places and don't stay too long. A camera monitors road signs and lane rules. If the driver skips stops or breaks traffic rules, the system logs the event. If there are several violations in a short time, it automatically sends a report to the authorities.

3. **Footboard and Overcrowding Alerts:** Cameras or sensors detect if passengers are standing on the footboard while the bus is moving. If so, the driver is warned. An automatic passenger counter checks for overcrowding. If the bus exceeds its limit, a warning is given and repeated issues are reported. Passengers can also send complaints using a QR code inside the bus.

4. **Rainy Weather & Smoking Alerts:** A rain sensor or weather forecast checks for rain, then compares it with the bus speed. If the bus is too fast in rainy or low-visibility conditions, the system warns the driver. Smoke and heat sensors also alert the driver if someone smokes or lights a cigarette inside.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Our Bus Passenger Safety System has four main parts, and each needs special knowledge, proper data, and affordable tools to work.

Driver Fatigue Detection – We need to know computer vision and machine learning. We can use tools like MediaPipe or OpenCV to detect closed eyes, yawning, or head nodding. GPS and timers help track driving hours. We can collect real-time video of the driver using a low-cost camera (like USB webcam or Pi Cam), and simulate fatigue cases to train a basic model. As software students, we can write Python code to process video and trigger alerts using a buzzer.

Route and Road Violation Detection – We must understand GPS tracking, geofencing, and local road rules. We can use Google Maps API to get bus stop locations and create geofenced zones. With a camera and OpenCV, we can detect traffic signs. Data like GPS logs and timestamps can be tested using Arduino + GPS module. We can use Python or Node.js to send violation logs to cloud databases.

Footboard and Overcrowding Alerts – Using a simple IR sensor or camera, we can detect people standing on the footboard. For passenger counting, we can try cheap image-processing methods or motion sensors. For feedback, we can design a simple mobile app or use Google Forms with QR codes. As students, we can build these systems with Raspberry Pi or ESP32, and code in Arduino or Python.

Rainy Weather & Smoking Alerts - We can use low-cost rain, smoke, and heat sensor modules connected to a microcontroller. When rain is predicted using a weather API or detected by the sensor, and the bus is speeding (via GPS), we alert the driver with a warning beep or display. Smoke or lighter heat detection also triggers alerts. We write the logic in C++ (Arduino) or Python to process sensor inputs, track speed, fetch forecasts, and manage real-time alerts effectively.

8. Objectives and Novelty

Main Objective The main objective of our project is to build a smart and low-cost bus passenger safety system that helps reduce accidents and improve rule-following in public transport, specially in Sri Lanka.We want to use simple sensors, cameras, GPS, and software code to detect problems like driver tiredness, skipping bus stops, footboard riding, overspeeding during rain, and even smoking inside the bus. Our goal is to warn the driver in real-time and also collect useful data that can be sent to bus companies or authorities. This system can help save lives, improve safety for school children and daily passengers, and make bus services more reliable.As software engineering students, we are focusing on writing code to connect all these devices, process real-time data, and make the system smart but affordable. Our main aim is to solve real problems in our country using the skills we learned in software engineering.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22030030 - Sadan M.D.T	**Driver Fatigue Detection System**	I will use a camera to check the driver's face to see if they are sleepy. I will write code to detect if eyes are closing, yawning, or head is nodding. I will also use GPS to check how long the driver has been driving.	We are not just checking the face — we also use the SLTB rule about 6-hour drive time, and combine both to alert the driver. This makes our system more useful than just a normal drowsiness detector.
IT22886286 - De Silva U.M.T.K	**Route and Road Violation Detection System**	I will use GPS to make sure the bus stops at the right places. I will also use a camera to read road signs like "Stop" or "No Entry" and warn the driver if they break the rule.	Our system works during the whole trip, not just at checkpoints. It watches the road and the GPS all the time and automatically records if something is wrong.

IT22142696 - Dias D.G.M	**Footboard and Overcrowding Alerts System**	I will set up a small sensor or camera near the bus door to check if anyone is standing on the footboard for too long. I will also make a simple app or QR form so passengers can report issues like speeding or overcrowding.	No one has added a system to warn about footboard riding before. And our idea to let passengers send complaints directly is simple but new for local buses.
IT22356154 - Gamlath G.R.A.A	**Rainy Weather & Smoking Alerts System**	I will use a rain sensor to detect rain and check speed with GPS. If the bus is too fast, it will beep. I'll also add a smoke and heat sensor to warn if someone smokes. A weather API warns before rain. The system suggests safe speeds based on rain, traffic, and visibility.	We mix early weather alerts, advisory speed, and smoke-heat detection in one cheap system. These are usually separate, but we combine them with simple sensors, a forecast API, and basic logic

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22030030 – Sadan M.D.T	This part uses computer vision, which is a software technique we learned through OpenCV and AI tools. We write code to detect face landmarks (eyes, mouth, head position) using Python. We also use GPS data and timers, and connect everything through code that checks driving time and alerts the driver. This includes real-time processing, sensor integration, and logic control, which are key areas in software engineering.
IT22886286 – De Silva U.M.T.K	We use GPS and geofencing, which are common in mobile and location-based software systems. Writing code to check if the bus is on the correct path and logging violations teaches us how to handle data streams, APIs, and maps. We also use camera-based road sign detection using image processing, which involves machine learning models, data annotation, and code testing, all part of software development.
IT22142696 - Dias D.G.M	Here, we write logic to detect passengers standing on the footboard using sensors or cameras. We also design a simple mobile app or complaint system using UI/UX design, backend logic, and feedback handling. These tasks are directly related to full-stack development, real-time software, and user interaction design, all covered in our software courses.
IT22356154 - Gamlath G.R.A.A	We connect sensors (rain, smoke, heat) to a microcontroller and write embedded software to process inputs. We also fetch weather data to predict rain and suggest safe speeds. This teaches sensor fusion, IoT logic, and reliable embedded coding, key skills in software engineering.

10. Supervisor details

	Title	First Name	Last Name	Signature

Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.